

Lab 6: Designing Jump and Bounce

Motivations

In this lab you will practice to

- avoid hardcoding, i.e. guessing all types of magic numbers that are the foundation of simulation; in our case the game play.
- design motion trajectory on *paper*, sort of: calibration step
- derive values/constants to model that particular motion in code

Jumps

Make a copy of your lab04, task 2, i.e. your jump.

Using the theory we saw in class, replicate the jump calibration of the provided two videos.

With respect to time: **C** key

To start, you should use a fixed time, like 0.5 second. Setting a user height marker, values are derived so your character reaches that max height using a parabolic motion.

Check that part with me before continuing.

When that is working, you should add keys so the fixed time can be updated as in the video.

With respect to lateral motion: **X** key

Now you enable a jump calibration in both directions.

The user marker provides also the lateral distance to reach, as in the second video.

Check that part with me before continuing.

Bounce

Check that part with me before continuing.

References

- [Kinematic jump formula](#)
- [Reverse Engineering a Jump](#)
- [Math for Game Programmers: Building a Better Jump](#) J. Kyle Pittman, @PirateHearts Co-founder, Minor Key Game [slides](#)
- [Coefficient of Restitution](#)
- [Bouncing Ball Physics](#)

Submission

- Code for both task: include both names at top of the file containing most of the code.

Planning: Fill-in below

Name: _____ Partner: _____

Roadmap

For each of the three parts, describe how are you processing after discussing with your partner.

In particular, specify what are you using from lectures/references, equations you are using to implement a specific objective.